NeuroFit Headband

Mind and Health Tracker Device

Project Documentation

1. Introduction

The **Mind and Health Tracker** is an advanced wearable device that monitors the user's mental and physical health in real time. The device integrates key health sensors in a headband form factor and transmits data via Bluetooth Low Energy (BLE) to a web-based application. The application not only visualizes this data but also leverages machine learning to predict the user's mood and offers guided mental wellness support. The system includes automatic emergency alerts based on abnormal readings.

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2. System Overview

Key Features:

- Real-time monitoring of brain activity, heart rate, and SpO2.
- BLE-based data transfer to a secure web application.
- Mood prediction using machine learning.

- Guided meditation sessions based on health analytics.
- Emergency alert system with GPS location sharing.

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3. Hardware Components

1. Seed Studio Xiao nRF52840

- Microcontroller handling sensor inputs, BLE communication, and data transmission.

2. BioAmp EXG Pill

- EEG sensor for tracking brain wave activity.
- Analyzes Delta, Theta, Alpha, Beta, and Gamma frequencies.

3. MAX30102 Pulse Oximeter

- Measures heart rate and oxygen saturation (SpO2).

4. Vibration Motor

- Provides haptic feedback for notifications (e.g., emergency or meditation prompts).

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4. Brain Waves: Classification and Significance

The brain communicates via electrical impulses. These impulses occur at various frequencies and are classified into five main types of brain waves. Understanding these waves helps determine the user's mental state.

A. Delta Waves (0.5 – 4 Hz)

- **Associated State:** Deep sleep, unconsciousness
- **Function:** Body healing, regeneration
- **Indicators:** High delta activity can indicate restorative sleep or potential neurological issues if seen during wakefulness

B. Theta Waves (4 – 8 Hz)

- **Associated State:** Light sleep, deep meditation, creativity
- **Function:** Subconscious insight, memory access

- **Indicators:** Increased theta is linked to creativity but may indicate inattention in active settings

C. Alpha Waves (8 – 12 Hz)

- **Associated State:** Relaxed alertness, calm focus
- **Function:** Stress reduction, coordination
- **Indicators:** Balanced alpha activity is ideal for relaxed mental states

D. Beta Waves (12 – 30 Hz)

- **Associated State:** Active thinking, problem-solving, alertness
- **Function:** Cognitive tasks, focus, logical reasoning
- **Indicators:** High beta may indicate stress or anxiety

E. Gamma Waves (30 – 100 Hz)

- **Associated State:** High-level information processing, learning
- **Function:** Memory retention, perception
- **Indicators:** High gamma levels correlate with cognitive sharpness

The EEG sensor (BioAmp EXG Pill) will measure these frequencies, and the system will classify mental states accordingly. Mood prediction and guided meditation are aligned with this analysis.

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5. Software Architecture

A. Frontend (Web Application)

- **Technologies:** HTML, CSS, JavaScript, Bootstrap
- **Libraries:** Chart.js (for visualization), Socket.io (for real-time updates if needed)

Features:

- **Dashboard** displaying live heart rate, SpO2, and brainwave data.
- **Mood Score Panel** updated in real time using ML output.
- **Historical Analysis** to view trends over time.
- **Meditation Guide** recommendations based on current stress levels.

B. Backend (Django Framework)

- **Functions:** BLE data reception, user authentication, database operations, ML processing
- **Database:** PostgreSQL or MySQL

Modules:

- **Data Ingestion**: Receives BLE data and logs it.
- **Authentication**: Manages secure sign-ups/logins.
- **Mood Prediction Engine**: Uses a trained ML model (TensorFlow or scikit-learn).
- **Emergency System**: Detects anomalies and triggers alerts with GPS data.

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6. Machine Learning Model

Objective:

Predict user's mood (e.g., Calm, Stressed, Focused, Happy) based on EEG, heart rate, and SpO2 data.

Model Details:

- **Type**: Supervised Learning
- **Input**: Brain wave frequency bands, HRV, SpO2
- **Output**: Mood class/score
- **Tools**: TensorFlow Lite (for embedded), Python (for backend training)
- **Training Dataset**: Pre-collected health data labeled with mood states

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7. System Workflow

1. **Data Collection:**
    - Brainwave (EEG), heart rate, and SpO2 are recorded continuously by sensors.
2. **BLE Transmission:**
    - The Xiao nRF52840 sends data via BLE to the frontend or backend.
3. **Backend Processing:**

- Data is stored in the database.
- Analyzed by ML models.
- Emergency triggers checked.

4. **Frontend Visualization:**
   - Data visualized with real-time charts.
   - Mood predictions and meditation guides updated.
5. **Emergency Handling:**
   - Abnormal patterns (e.g., low SpO2 or EEG spike) trigger alerts.
   - Location and status sent to emergency contact.

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8. Features Summary

Feature	Description
Sign-Up/Login	Secure user authentication.
Dashboard	Real-time health data visualization.
Historical Analytics	Filter and view past data trends.
Mood Prediction	AI-based mood scoring.
Meditation Mode	Guided relaxation sessions.
Emergency Alerts	GPS-linked SMS/email alerts.

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9. Technology Stack

Component	Technology
Microcontroller	Seed Studio Xiao nRF52840
EEG Sensor	BioAmp EXG Pill
Pulse Sensor	MAX30102
Frontend	HTML, CSS, JS, Chart.js, Bootstrap
Backend	Django, PostgreSQL/MySQL
ML Model	TensorFlow, Scikit-learn

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10. Project Timeline

Milestone	Timeframe
Requirements & Design	Week 1-2
Hardware Integration	Week 3-5
Frontend Development	Week 6-8
Backend Development	Week 7-9
ML Model Integration	Week 8-10
Testing & Debugging	Week 10-12
Final Deployment	Week 13

11. Conclusion

The Mind and Health Tracker is designed to be a powerful, compact, and user-friendly solution for mental and physical health monitoring. It combines sensor technology, BLE communication, web development, and machine learning to provide real-time feedback and actionable health insights. This innovation will be valuable not only to individual users but also in healthcare and wellness research.

By providing real-time analysis of brain waves and other physiological signals, this system gives users actionable insights into their mental state. With emergency alert systems and personalized meditation guides, it moves beyond monitoring to actively supporting well-being and safety.

Github Repository Structure (Sample)

```
NeuroFit/
├── Arduino/
│   ├── MindHealthTracker.ino
│   ├── config.h
│   ├── BLEService.h
│   ├── heartRate.h
│   └── spo2_algorithm.c
│
├── FlutterApp/ (optional if using Flutter frontend)
│   ├── lib/
```

```
│   │   ├── main.dart
│   │   ├── screens/
│   │   ├── services/
│   │   └── widgets/
│   ├── pubspec.yaml
│   └── android/app/src/main/AndroidManifest.xml
│
├── WebApp/
│   ├── index.html
│   ├── style.css
│   ├── app.js
│   └── assets/
│       └── charts.js
│
├── Backend/
│   ├── manage.py
│   ├── tracker/
│   │   ├── models.py
│   │   ├── views.py
│   │   ├── urls.py
│   │   └── ble_receiver.py
│   ├── templates/
│   └── static/
│
├── ML_Model/
│   ├── train_model.ipynb
│   ├── mood_classifier.tflite
│   └── utils.py
│
├── Hardware_Schematics/
│   ├── schematic.png
│   ├── wiring_diagram.pdf
│   └── BOM.csv
│
├── README.md
├── LICENSE
└── .gitignore
```

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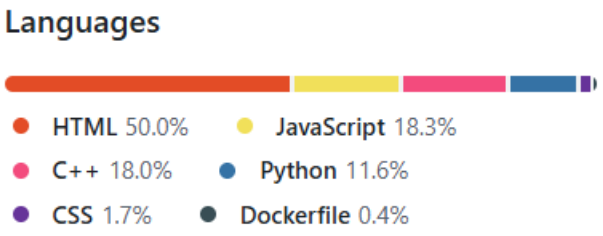
Progress report date wise

Date - 27/07/2025

Entry- Completed the basic frontend and backend (with dashboard, health data, meditation page, and emergency email feature)

Github repository - https://github.com/anu-rag-panda/neurofit-v1.0.1/

language stack till now -



Date- 28/07/2025

Entry- Successfully compiled the basic hardware code

Hardware Code

Date- 29/07/2025

Entry- Implemented the emergency SMS feature

Emergency SMS feature